Activity: nested hypothesis tests

Group members:

In this class activity, we will continue working with the `Kids198` data from the `Stat2Data` package, which examines the relationship between Age and Height for children between the ages of 8 and 18. We are also interested in how that relationship differs for boys and girls. The `Kids198` dataset includes the following variables on 198 children:

- `Height`: Height (in inches)
- `Age`: Age (in months)
- `Sex`: 0=male, 1 = female

Questions

Researchers will treat Height as the response, and Age and Sex as the predictors. They want to know whether the slope on Age (i.e., the increase in Height associated with a one-month increase in Age) is the same for boys and girls.

1. What model should the researchers use to test this research question?

2. What are the full and reduced models that the researchers will compare to test their question?

Now the researchers want to know whether there is *any* relationship between Age and Height, after accounting for Sex. They want to test this relationship, using the interaction model from exercise 1 (in case the relationship is different for boys and girls).

3. What are the full and reduced models that the researchers will compare to test their new question?